

Linear algebra – 4

Functions of matrices and matrix equations

12 сентября 2026 года

Theoretical minimum.

Theorem 1 (Cayley–Hamilton). $\chi_A(A) = 0$. Consequently every power A^m with $m \geq n$ is a combination of $E, A, \dots, A^{n-1}$, an invertible A has A^{-1} equal to a polynomial in A , and any identity to be proved for all powers need only be checked on the first n .

Definition (functions of a matrix). For f analytic near the spectrum, $f(A)$ is defined blockwise on the Jordan form by $f(J_k(\lambda)) = \sum_{j < k} \frac{f^{(j)}(\lambda)}{j!} N^j$, equivalently by $f(A) = \frac{1}{2\pi i} \oint f(z)(zE - A)^{-1} dz$. In particular $e^A = \sum_{k \geq 0} A^k/k!$ and $\text{Spec } f(A) = f(\text{Spec } A)$; the value of $f(A)$ depends only on the values of f and of finitely many of its derivatives at the eigenvalues.

Theorem 2 (Sylvester’s equation). $AX - XB = C$ has a unique solution for every C if and only if A and B have no common eigenvalue. In the language of the Kronecker product $\text{vec}(AXB) = (B^\top \otimes A) \text{vec } X$, and the spectrum of $A \otimes E - E \otimes B^\top$ is $\{\lambda_i - \mu_j\}$ — which is exactly the criterion.

Fact 1 (non-commutativity is the whole difficulty). $e^A e^B = e^{A+B}$ requires $AB = BA$; in general only the Lie–Trotter formula $e^{A+B} = \lim_{n \rightarrow \infty} (e^{A/n} e^{B/n})^n$ holds. Still, AB and BA always have the same characteristic polynomial, so spectral statements about one transfer to the other.

Fact 2 (solving a matrix equation). Three standard moves: pass to the Jordan form and solve blockwise; read the equation as a linear map on M_n and compute its spectrum through the Kronecker product; or apply the equation to an eigenvector and reduce to numbers. If the equation involves a square root or a logarithm, the answer is governed by the sizes of the Jordan blocks.

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1. Prove the criterion for unique solvability of $AX - XB = C$, both through the Kronecker product and through the integral $X = \frac{1}{2\pi i} \oint (zE - A)^{-1} C (zE - B)^{-1} dz$; deduce that if $AX = XA$ for all X , then A is scalar.
 2. (VJIMC 2005) For an arbitrary square matrix M define $\exp(M) = E + \frac{M}{1!} + \frac{M^2}{2!} + \frac{M^3}{3!} + \dots$. Construct 2×2 matrices A and B with $\exp(A + B) \neq \exp(A)\exp(B)$.
 3. (Putnam 1996) For a square matrix A define $\sin A = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} A^{2n+1}$. Prove or disprove: there exists a 2×2 matrix A with real entries such that $\sin A = \begin{pmatrix} 1 & 1996 \\ 0 & 1 \end{pmatrix}$.
 4. (IMC 2000) For a real $m \times m$ matrix A let $e^A = \sum_{n \geq 0} \frac{1}{n!} A^n$. Prove or disprove: for all real polynomials p and all real $m \times m$ matrices A and B , the matrix $p(e^{AB})$ is nilpotent if and only if $p(e^{BA})$ is nilpotent.
 5. (IMC) Let A be a real $n \times n$ matrix with $A^3 = O$.
 - a) Prove that there is a unique real X with $X + AX + XA^2 = A$.
 - b) Express X in terms of A .
 6. (IMC 2025 shortlist) Let A be a square matrix over a field F with $A^n = O$. Prove that the equation

$$\sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \alpha_{ij} A^i X A^j = B, \quad \alpha_{ij} \in F, \alpha_{00} \neq 0,$$

is solvable for every square matrix B of the same size.

7. (ФКН, весна 2024) Find all matrices X with $X^2 = A$, where

$$A = \begin{pmatrix} 0.25 & 1 & 0 & 0 \\ 0 & 0.25 & 0 & 0 \\ 0 & 0 & 0.25 & 1 \\ 0 & 0 & 0 & 0.25 \end{pmatrix}.$$